

At Least One of $\zeta(7), \zeta(9), \dots, \zeta(19)$ Is Irrational

A Growing-Determinant Proof with Lean Certification

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Abstract

We prove that at least one of the seven odd zeta values from $\zeta(7)$ through $\zeta(19)$ is irrational. A higher-derivative rational kernel gives a sequence of $2n \times 2n$ matrices whose entries are affine rational combinations of these values. Primitive normalization converts their determinants into integer polynomials of degree at most $2n$. We establish the global estimate $|\det H_n/c_n| \leq \exp(-n^2/2)$ for all sufficiently large n . A reduction at primes $55n + 1$ proves nonvanishing under the joint-rationality assumption, completing the integer contradiction. The proof is formalized in Lean using Mathlib's Riemann zeta function. This article gives the construction, the quantitative assembly, and the correspondence with the complete machine-checked proof distributed with this version.

Keywords. Odd zeta values; irrationality; hypergeometric forms; determinants; formal verification.

Mathematics Subject Classification. 11J72, 11M06, 68V15.

1 The seven-value theorem

For integers $s > 1$, write $\zeta(s) = \sum_{k=1}^{\infty} k^{-s}$. Our main result is the following.

Theorem 1.1. *At least one of $\zeta(7), \zeta(9), \zeta(11), \zeta(13), \zeta(15), \zeta(17), \zeta(19)$ is irrational.*

The construction belongs to the higher-derivative hypergeometric framework developed by Zudilin [1]. Public proof developments of Devincenzis [2] and CoolRmal [3] establish the eight-value result ending at $\zeta(21)$. Theorem 1.1 shortens that list by one value.

The determinant, rather than an individual linear form, is the decisive object here. Its size grows with n , so both the analytic estimate and the arithmetic cost occur on the n^2 scale. Under joint rationality, clearing the seven fixed denominators costs only Q^{2n} . The resulting quadratic-exponential decay contradicts nonzero integrality.

We use the 158-centered staircase kernel described below. This parameter family and related local denominator refinements occur in the development notes associated with [2]. The contribution of the present proof is the growing-determinant construction and its completed arithmetic, analytic, and nonvanishing estimates for the seven-value list. The attached Lean development supplies the detailed proofs of the estimates used in Sections 3 and 4; Section 6 identifies their exact declarations.

2 The rational kernel and matrix

For $m \geq 0$, let $(x)_m = \prod_{j=0}^{m-1} (x+j)$. Define

$$\kappa_n = 2 \frac{\prod_{b=0}^{15} ((52-2b)n)!}{\prod_{a=0}^4 ((48+a)n)!^2}, \quad (1)$$

$$R_n(t) = \kappa_n (t+79n+1) \frac{\prod_{a=0}^4 (t+1)_{(48+a)n} (t+158n+2-(48+a)n)_{(48+a)n}}{\prod_{b=0}^{15} (t+(53+b)n+1)_{(52-2b)n+1}}. \quad (2)$$

Its degree at infinity is $-92n-15$. The numerator vanishes to order at least five at every $-k$ with $1 \leq k \leq 48n$, and the denominator is nonzero at those points.

Use the even, integer-valued polynomial basis

$$E_0(x) = 1, \quad E_i(x) = \frac{2x^2 \prod_{j=1}^{i-1} (x^2 - j^2)}{(2i)!} \quad (i \geq 1). \quad (3)$$

For $i \geq 1$ the identity $E_i(x) = \binom{x+i}{2i} + \binom{x+i-1}{2i}$ proves integer-valuedness on $\mathbb{Z}$. Set $d = 2n$ and, for $0 \leq i, j < d$, put

$$R_{n,i,j}(t) = R_n(t) E_i(t+79n+1) E_j(t+79n+1), \quad (4)$$

$$H_n(i, j) = \frac{1}{4!} \sum_{v=0}^{\infty} R_{n,i,j}^{(4)}(v-48n). \quad (5)$$

Every entry kernel has degree at infinity at most $-84n-19$. The fourth-derivative series in (5) therefore converges absolutely. All its evaluation points lie strictly to the right of its poles.

2.1 The seven coordinates

Expand the entry kernel in its unique partial fractions,

$$R_{n,i,j}(t) = \sum_k \sum_{\ell=1}^{m_{n,k}} \frac{a_{n,i,j,k,\ell}}{(t+k)^\ell}, \quad m_{n,k} \leq 16, \quad (6)$$

with rational coefficients. The sums over k are finite and supported between $53n+1$ and $105n+1$. Write $h_s(K) = \sum_{v=1}^K v^{-s}$. Termwise differentiation and summation give

$$H_n(i, j) = \sum_k \sum_{\ell=1}^{m_{n,k}} a_{n,i,j,k,\ell} \binom{\ell+3}{4} (\zeta(\ell+4) - h_{\ell+4}(k-48n-1)). \quad (7)$$

The reflection $t \mapsto -t-158n-2$ makes the entry kernel odd. Pairing reflected poles cancels the even zeta coordinates. The degree at infinity is at most -2 , which gives $\sum_k a_{n,i,j,k,1} = 0$, cancelling the $\zeta(5)$ coordinate. Consequently

$$H_n(i, j) = b_{n,i,j} + \sum_{r=0}^6 b_{n,i,j,r} \zeta(7+2r), \quad b_{n,i,j}, b_{n,i,j,r} \in \mathbb{Q}. \quad (8)$$

The constant in (8) includes the entire harmonic correction in (7).

Let $\mathcal{H}_n(\mathbf{X})$ denote the matrix obtained from (8) by replacing $\zeta(7+2r)$ with X_r , and set $D_n(\mathbf{X}) = \det \mathcal{H}_n(\mathbf{X})$. This is a rational polynomial in seven variables of total degree at most $2n$. For $\mathbf{z} = (\zeta(7), \dots, \zeta(19))$, the exact identity is

$$D_n(\mathbf{z}) = \det H_n. \quad (9)$$

3 Primitive normalization and the global estimate

When $D_n \neq 0$, choose $c_n \in \mathbb{Q}^\times$ and $P_n \in \mathbb{Z}[X_0, \dots, X_6]$ with content one such that

$$D_n = c_n P_n, \quad \deg P_n \leq 2n. \quad (10)$$

For the zero polynomial use $c_n = 1$ and $P_n = 0$. Thus $c_n \neq 0$ for every n , and

$$P_n(\mathbf{z}) = \frac{\det H_n}{c_n} \quad (11)$$

holds for every index. Define the logarithmic arithmetic cost $C_n = -\log |c_n|$ and the constant

$$A = \frac{8123483}{8000} - 12 + 8 \log 4. \quad (12)$$

Proposition 3.1 (Certified arithmetic and analytic estimates). *For every $\epsilon > 0$, the following inequalities hold for all sufficiently large n :*

$$C_n \leq (1012 + \epsilon)n^2 \quad \text{if } D_n \neq 0, \quad (13)$$

$$|\det H_n| \leq \exp((-A + 2 + \epsilon)n^2). \quad (14)$$

Moreover $A > 1014 + \frac{1}{2}$.

The arithmetic estimate uses the prime-by-prime Gauss valuations of the same polynomial D_n . Its three certified contributions have upper quadratic rates 74, 433, and 505, which sum to 1012. The assembly retains exceptional primes and endpoint corrections. The passage from prime sums to rates uses the bundled formal proof of the prime number theorem.

For the analytic estimate, a change to a scaled monic even basis preserves the determinant up to its explicit leading-coefficient normalization. Bounds on modulus moments give a product estimate over the $2n$ rows. The compact phase bounds, the infinite tails, and finite-index drift are included in that estimate. A uniform allowance of one per row contributes the $+2$ on the quadratic scale. The determinant's factorial prefactor and polynomial prefactors contribute $o(n^2)$; the exact basis normalization contributes the remaining terms in (12).

These statements are proved by the declarations `h158ArithmeticCertifiedRate` and `h158AnalyticCertifiedRate`. Their source dependencies contain the finite rational certificates and their all-index assembly. The numerical constant comparison is the theorem `h158ConservativeBudget_slack_gap`.

Theorem 3.2 (Global determinant estimate). *For all sufficiently large integers n ,*

$$\left| \frac{\det H_n}{c_n} \right| = |P_n(\mathbf{z})| \leq e^{-n^2/2}. \quad (15)$$

Proof. If $D_n = 0$, identity (9) makes the left-hand side zero. Otherwise apply Proposition 3.1 with $\epsilon = (A - 1014 - \frac{1}{2})/3 > 0$. Since $|c_n|^{-1} = e^{C_n}$,

$$\left| \frac{\det H_n}{c_n} \right| \leq \exp((1014 - A + 2\epsilon)n^2) \leq \exp(-n^2/2).$$

Equation (11) supplies the polynomial formulation. □

4 Nonvanishing at auxiliary primes

The next argument supplies nonvanishing for rational specializations, precisely as needed in the proof by contradiction.

Proposition 4.1 (Auxiliary-prime nonvanishing). *Let $p = 55n + 1$ be prime, and let $\mathbf{x} \in \mathbb{Q}^7$ have coordinates integral at p . Then*

$$D_n(\mathbf{x}) \neq 0, \quad v_p(D_n(\mathbf{x})) = -4n.$$

The entrywise local computation proves that $p^2 \mathcal{H}_n(\mathbf{x})$ is integral at p and that its reduction satisfies

$$\overline{p^2 \mathcal{H}_n(\mathbf{x})} = V_n W_n V_n^\top \quad \text{over } \mathbb{F}_p, \quad (16)$$

where W_n is diagonal with every diagonal entry nonzero. The entries of V_n are the reductions of $E_i(u_j)$ at

$$u_j = 24n + 1 + j, \quad 0 \leq j < 2n.$$

The complete local reduction is established in `H158GateAssembly.lean`; its weights are defined there from the Laurent coefficients of the concrete kernel (2).

To see why this reduction proves the proposition, observe that $0 < u_j < p$ and $u_j + u_k < p$. Hence the squares u_j^2 are pairwise distinct modulo p . Each $E_i(u)$ is a degree- i polynomial in u^2 with nonzero leading coefficient modulo p , because $2i < p$. Thus V_n is an invertible change of basis of a Vandermonde matrix. Taking determinants in (16) shows that $p^2 \mathcal{H}_n(\mathbf{x})$ has unit determinant in the local ring. As its size is $2n$,

$$0 = v_p(\det(p^2 \mathcal{H}_n(\mathbf{x}))) = 4n + v_p(D_n(\mathbf{x})),$$

which proves both assertions.

5 Proof of Theorem 1.1

Suppose that all seven coordinates of $\mathbf{z}$ are rational. Choose a positive integer Q such that $Qz_r \in \mathbb{Z}$ for every $0 \leq r \leq 6$. By the degree bound in (10),

$$I_n = Q^{2n} P_n(\mathbf{z}) \in \mathbb{Z} \quad (17)$$

for every n . This follows monomial by monomial: a monomial of degree $k \leq 2n$ acquires an integral coefficient after multiplication by Q^{2n} .

Dirichlet's theorem on primes in arithmetic progressions gives infinitely many primes $p \equiv 1 \pmod{55}$. Omitting the finitely many prime divisors of Q leaves an unbounded set of indices n with $p = 55n + 1$ prime and $p \nmid Q$. At each such index the coordinates of $\mathbf{z}$ are integral at p , and Proposition 4.1 gives $D_n(\mathbf{z}) \neq 0$. Equations (10) and (17) therefore give $I_n \neq 0$.

On this same unbounded set, Theorem 3.2 implies

$$0 < |I_n| \leq \exp(2n \log Q - n^2/2) < 1$$

for all sufficiently large n . This is impossible for an integer. Theorem 1.1 follows. $\square$

6 Lean certification and reproducibility

The final declaration is

```
OddZetaMixed.exists_irrational_riemannZeta_seven_to_nineteen.
```

It asserts the existence of an exponent in $\{7, 9, 11, 13, 15, 17, 19\}$ for which the real part of Mathlib’s `riemannZeta` is irrational. At these real arguments the complex Riemann zeta function equals its real Dirichlet series. The declaration has no additional mathematical hypotheses. Its transitive axiom dependencies are `propext`, `Classical.choice`, and `Quot.sound`.

Proof component	Source in OddZetaMixed/
Kernel and basis	H158Kernel.lean, EvenBasis.lean, KernelBasis.lean
Convergent matrix entries	H158Functional.lean
Seven-coordinate identity	H158SevenIdentity.lean
Primitive normalization	H158Normalization.lean
Arithmetic rate	H158ArithmeticCertified.lean
Analytic rate	H158AnalyticCertified.lean
Auxiliary-prime reduction	H158GateAssembly.lean, GateVandermonde.lean
Global bound and theorem	H158SevenIrrational.lean

The accepted source replay recorded on 6 October 2026 passed all 300 checks, rebuilt 256 project modules, and recorded hashes for 326 proof, configuration, and verifier inputs. The bundled prime-number-theorem replay rebuilt 22 dependency modules and two audit files and checked 72 targets. These receipts accompany the source in `audit/2026-10-06/`.

The pinned toolchain is Lean v4.34.0-rc2, with Mathlib revision

```
dc4b8d60d5edb3c493c3662126b1b7ccae7d67cf.
```

The prime number theorem dependency is selected from the repository

```
AlexKontorovich/PrimeNumberTheoremAnd,
```

at revision

```
c39a751132c88b6e8080b74c74023fd95b3d8be0.
```

The repository records the source selection and compatibility transformations. The verifier recompiles project and bundled dependency sources while reusing the pinned foundational library artifacts.

From a checkout with the stated toolchain, run:

```
lake exe cache get
python3 tools/check_snapshot.py
python3 -m unittest test_verify
python3 verify.py
```

The snapshot check establishes byte identity with the recorded proof inputs; the full verifier performs the Lean source replay. The formal proof and certificate data are part of this version’s research artifact.

Availability and acknowledgments

The manuscript source, Lean development, dependency provenance, and audit receipts are distributed at <https://github.com/AlexxxxxLiu/odd-zeta-seven-lean>. This version is identified by the release tag `v1.0.0` and its accompanying SHA-256 inventory. The archival DOI is [10.5281/zenodo.23191331](https://doi.org/10.5281/zenodo.23191331). The manuscript is licensed under CC BY 4.0; original code is licensed under Apache-2.0, with upstream notices retained.

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References

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